

**Automated  
USB Data Loss  
Prevention (DLP)**  
using **Wazu**  **WAZU**

 Prevent Unauthorized USB Access

 Monitor & Scan USB Data

 Automate DLP Policies

 Ensure Compliance & Data Security

**SECURITY OPERATIONS CENTER**

THREAT INTELLIGENCE

78% SECURE

INCIDENTS

LIVE THREAT FEED

ENDPOINT PROTECTION

STATUS: PROTECTED

The background features a futuristic Security Operations Center with multiple large digital screens displaying various security metrics and maps. In the center, a glowing blue shield is superimposed over a laptop screen. To the right, a red USB drive with a skull and crossbones icon is shown with red energy trails connecting it to the shield, symbolizing data protection. The overall color scheme is dark blue and black with glowing blue and red highlights.

**Presented By:** Md. Kawsar Hossain  
**Date:** 27/9/2026

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## Executive Summary

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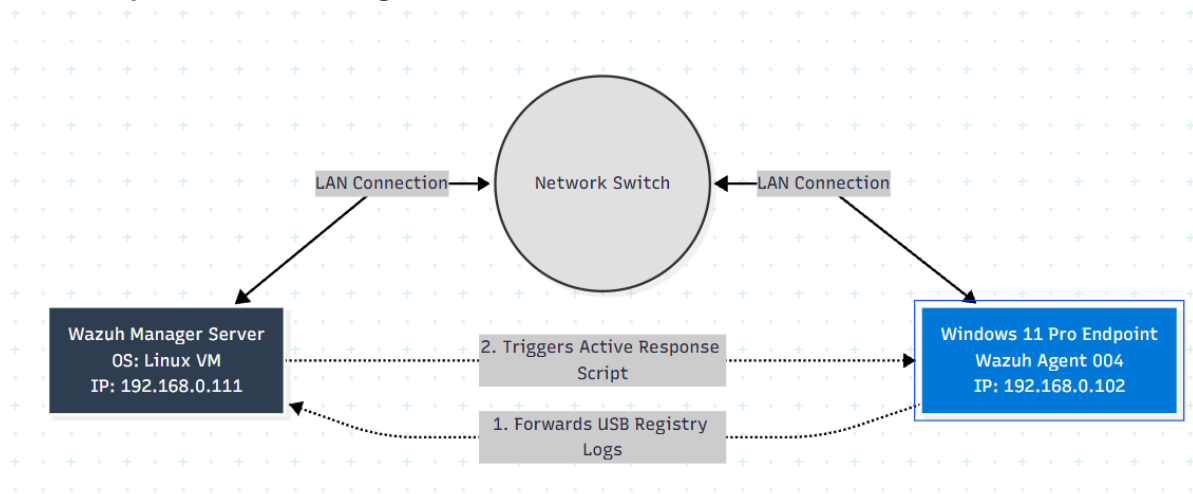
This project demonstrates the implementation of an automated Endpoint Threat Containment and Data Loss Prevention (DLP) pipeline using the Wazuh XDR/SIEM platform and custom Python-based Security Orchestration, Automation, and Response (SOAR) mechanisms.

Operating within a controlled environment comprising a Wazuh Manager (Linux VM) and a Windows 11 Pro endpoint, the architecture successfully detects unauthorized mass storage device attachments via granular Windows Registry monitoring (Rule 100016). Upon detection, the system triggers a stateful automated active response script (`usb_prevent_ar_script.exe`) that programmatically disables the USBSTOR driver to instantly neutralize potential hardware-based data exfiltration vectors, followed by an automated 180-second timeout recovery.

Deliverables include structured telemetry logs, automated active-response execution timelines, and zero-analyst-intervention hardware containment, significantly reducing Mean Time to Respond (MTTR) and enhancing enterprise endpoint security posture.

## Architecture and Setup:

- Soc Analysis Network Design:



This SOC analysis network design illustrates the communication architecture between the Wazuh Manager (Linux VM) and the Windows 11 Pro endpoint over the local network (LAN). It highlights the continuous forwarding of USB registry events from the agent, and the automated Active Response execution flow triggered by the Manager for real-time hardware containment.

## Agent Summary & Environment Overview:

| Agents (1)                                                                                                                                                              |      |               |               |                    |                                                                      |              |         |                               |                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------|---------------|--------------------|----------------------------------------------------------------------|--------------|---------|-------------------------------|---------------------------------------------------|
| <div><div>+</div> Deploy new agent</div> <div><div>↺</div> Refresh</div> <div><div>📄</div> Export formatted</div> <div>More <div>▼</div></div> <div><div>⚙️</div></div> |      |               |               |                    |                                                                      |              |         |                               |                                                   |
| <div>status=active</div>                                                                                                                                                |      |               |               |                    |                                                                      |              |         |                               | <div>WQL</div>                                    |
| <input type="checkbox"/>                                                                                                                                                | ID ↑ | Name          | IP address    | Group(s)           | Operating system                                                     | Cluster node | Version | Status                        | Actions                                           |
| <input type="checkbox"/>                                                                                                                                                | 004  | Windows_11pro | 192.168.0.102 | <div>default</div> | <div><div>🖥️</div>Microsoft Windows 11 Pro<br/>10.0.26100.4349</div> | node01       | v4.14.4 | <div><div>●</div>active</div> | <div><div>ℹ️</div><div>👁️</div><div>⋮</div></div> |

The Wazuh dashboard agents view confirms the successful deployment, registration, and active status of the endpoint. The target system is configured under Agent ID **004**, running **Microsoft Windows 11 Pro** (Build 10.0.26100.4349) with Wazuh agent version **v4.14.4** communicating securely with the cluster node. This active channel forms the foundation for real-time hardware telemetry and automated Active Response enforcement.

## Detection of Unauthorized USB Device:

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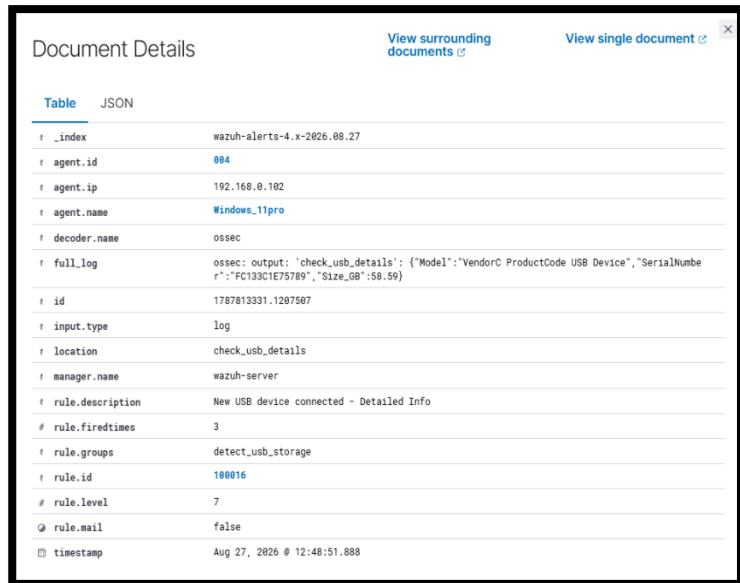
### *Proof of concept*

|                                                                                  |                              |               |                                          |   |        |
|----------------------------------------------------------------------------------|------------------------------|---------------|------------------------------------------|---|--------|
|  | Aug 27, 2026 @ 12:48:51.8... | Windows_11pro | New USB device connected - Detailed Info | 7 | 100016 |
|----------------------------------------------------------------------------------|------------------------------|---------------|------------------------------------------|---|--------|

The Wazuh dashboard captures the connection of a mass storage device on endpoint Windows\_11pro (Agent ID **004**). Rule **100016** (Level 7) successfully fires upon detecting the hardware insertion, logging precise telemetry details including device models and unique footprints.

# Granular USB Telemetry Extraction:

## Proof of concept



| Document Details                                                                |                                                                                                                                    |
|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">View surrounding documents</a> <a href="#">View single document</a> |                                                                                                                                    |
| Table                                                                           | JSON                                                                                                                               |
| † _index                                                                        | wazuh-alerts-4.x-2026.08.27                                                                                                        |
| † agent.id                                                                      | 004                                                                                                                                |
| † agent.ip                                                                      | 192.168.0.102                                                                                                                      |
| † agent.name                                                                    | Windows_11pro                                                                                                                      |
| † decoder.name                                                                  | ossec                                                                                                                              |
| † full_log                                                                      | ossec: output: 'check_usb_details': {'Model': 'VendorC ProductCode USB Device', 'SerialNumber': 'FC133C1E75789', 'Size_GB': 58.59} |
| † id                                                                            | 1787813331.1207507                                                                                                                 |
| † input.type                                                                    | log                                                                                                                                |
| † location                                                                      | check_usb_details                                                                                                                  |
| † manager.name                                                                  | wazuh-server                                                                                                                       |
| † rule.description                                                              | New USB device connected - Detailed Info                                                                                           |
| # rule.firedtimes                                                               | 3                                                                                                                                  |
| † rule.groups                                                                   | detect_usb_storage                                                                                                                 |
| † rule.id                                                                       | 100016                                                                                                                             |
| # rule.level                                                                    | 7                                                                                                                                  |
| @ rule.mail                                                                     | false                                                                                                                              |
| 🕒 timestamp                                                                     | Aug 27, 2026 @ 12:48:51.888                                                                                                        |

Inspecting the event JSON payload reveals deep log parsing capabilities. The system successfully extracts the device's hardware attributes—specifically the **Model**, exact **Serial Number** (FC133C1E75789), and **Storage Size** (58.59 GB). This structured data serves as the precise trigger condition for automated downstream containment.

# Automated Active Response Execution :

## Proof of concept.

| Document Details |                                        | <a href="#">View surrounding documents</a>                                                                                         | <a href="#">View single document</a> | ✕ |
|------------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|---|
| Table            | JSON                                   |                                                                                                                                    |                                      |   |
| ⌘                | _index                                 | wazuh-alerts-4.x-2026.08.27                                                                                                        |                                      |   |
| ⌘                | agent.id                               | 004                                                                                                                                |                                      |   |
| ⌘                | agent.ip                               | 192.168.0.102                                                                                                                      |                                      |   |
| ⌘                | agent.name                             | Windows_11pro                                                                                                                      |                                      |   |
| ⌘                | data.command                           | add                                                                                                                                |                                      |   |
| ⌘                | data.origin.module                     | wazuh-execd                                                                                                                        |                                      |   |
| ⌘                | data.origin.name                       | node01                                                                                                                             |                                      |   |
| ⌘                | data.parameters.alert.agent.id         | 004                                                                                                                                |                                      |   |
| ⌘                | data.parameters.alert.agent.ip         | 192.168.0.102                                                                                                                      |                                      |   |
| ⌘                | data.parameters.alert.agent.name       | Windows_11pro                                                                                                                      |                                      |   |
| ⌘                | data.parameters.alert.decoder.name     | ossec                                                                                                                              |                                      |   |
| ⌘                | data.parameters.alert.full_log         | ossec: output: 'check_usb_details': {"Model": "VendorC ProductCode USB Device", "SerialNumber": "FC133C1E75789", "Size_GB": 58.59} |                                      |   |
| ⌘                | data.parameters.alert.id               | 1787813331.1207507                                                                                                                 |                                      |   |
| ⌘                | data.parameters.alert.location         | check_usb_details                                                                                                                  |                                      |   |
| ⌘                | data.parameters.alert.manager.name     | wazuh-server                                                                                                                       |                                      |   |
| ⌘                | data.parameters.alert.rule.description | New USB device connected - Detailed Info                                                                                           |                                      |   |
| ⌘                | data.parameters.alert.rule.firedtimes  | 3                                                                                                                                  |                                      |   |
| ⌘                | data.parameters.alert.rule.groups      | detect_usb_storage                                                                                                                 |                                      |   |



# Automated Active Response Execution:

- *Proof of concept.*

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data.parameters.extra_args |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| data.parameters.program    | active-response/bin/usb_prevent_ar_script.exe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| data.version               | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| decoder.name               | ar_log_json                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| decoder.parent             | ar_log_json                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| full_log                   | 2026/08/27 12:48:51 active-response/bin/usb_prevent_ar_script.exe: {"version":1,"origin":{"name":"node01","module":"wazuh-execd"},"command":"add","parameters":{"extra_args":[],"alert":{"timestamp":"2026-08-27T06:48:51.888+0000"},"rule":{"level":7,"description":"New USB device connected - Detailed Info","id":"100016","firedtimes":3,"mail":false,"groups":["detect_usb_storage"]},"agent":{"id":"004","name":"Windows_11pro","ip":"192.168.0.102"},"manager":{"name":"wazuh-server"},"id":"1787813331.1207507","full_log":{"ossec":{"check_usb_details":{"Model":"VendorC ProductCode USB Device"},"SerialNumber":"5C133C15757801787813332.1207814"}} |
| id                         | 1787813332.1207814                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| input.type                 | log                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| location                   | active-response\active-responses.log                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| manager.name               | wazuh-server                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| rule.description           | Active response: active-response/bin/usb_prevent_ar_script.exe - add                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| rule.firedtimes            | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| rule.gdpr                  | IV_35.7.d                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| rule.groups                | ossec, active_response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| rule.id                    | 657                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| rule.level                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| rule.mail                  | false                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| rule.nist_800_53           | SI.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| rule.pci_dss               | 11.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| rule.tsc                   | CC6.1, CC6.8, CC7.2, CC7.3, CC7.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| timestamp                  | Aug 27, 2026 @ 12:48:52.855                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

Upon the successful triggering of detection Rule **100016**, the Wazuh Manager immediately initiates an automated remediation workflow via its wazuh-execd communication channel. A remote execution command is dispatched to the Windows 11 Pro endpoint (Agent**004**), invoking the custom stateful executable **usb\_prevent\_ar\_script.exe** residing in the agent's active-response directory. Without requiring any manual SOC analyst intervention, the script programmatically modifies the Windows Registry (USBSTOR start state) to instantly contain the hardware-based threat, effectively neutralizing potential data exfiltration vectors. This automated orchestration successfully bridges the gap between real-time threat detection and immediate containment, drastically reducing the Mean Time to Respond (MTTR)

## Stateful Containment & Event Timeline

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```
2026/08/27 12:48:51 active-response/bin/usb_prevent_ar_script.exe: {"version": 1, "origin": {"name": "active-response/bin/usb_prevent_ar_script.exe", "module": "active-response"}, "command": "check_keys", "parameters": {"keys": ["100016"]}}
2026/08/27 12:48:51 active-response/bin/usb_prevent_ar_script.exe: {"version":1,"origin":{"name":"node01","module":"wazuh-execd"},"command":"continue","parameters":{"extra_args":[],"alert":{"timestamp":"2026-08-27T06:48:51.888+0000","rule":{"level":7,"description":"New USB device connected - Detailed Info","id":"100016","firedtimes":3,"mail":false,"groups":["detect_usb_storage"]},"agent":{"id":"004","name":"Windows_11pro","ip":"192.168.0.102"},"manager":{"name":"wazuh-server"},"id":"1787813331.1207507"},"full_log":"ossec: output: 'check_usb_details': {\\\"Model\\\":\\\"VendorC ProductCode USB Device\\\",\\\"SerialNumber\\\":\\\"FC133C1E75789\\\",\\\"Size_GB\\\":58.59}}","decoder":{"name":"ossec"},"location":"check_usb_details"},"program":"active-response/bin/usb_prevent_ar_script.exe"}}
2026/08/27 12:48:51 active-response/bin/usb_prevent_ar_script.exe: USB storage disabled (rule ['100016'])
2026/08/27 12:48:51 active-response/bin/usb_prevent_ar_script.exe: Ended
2026/08/27 12:51:52 active-response/bin/usb_prevent_ar_script.exe: Started
2026/08/27 12:51:52 active-response/bin/usb_prevent_ar_script.exe: {"version":1,"origin":{"name":"node01","module":"wazuh-execd"},"command":"delete","parameters":{"extra_args":[],"alert":{"timestamp":"2026-08-27T06:48:51.888+0000","rule":{"level":7,"description":"New USB device connected - Detailed Info","id":"100016","firedtimes":3,"mail":false,"groups":["detect_usb_storage"]},"agent":{"id":"004","name":"Windows_11pro","ip":"192.168.0.102"},"manager":{"name":"wazuh-server"},"id":"1787813331.1207507"},"full_log":"ossec: output: 'check_usb_details': {\\\"Model\\\":\\\"VendorC ProductCode USB Device\\\",\\\"SerialNumber\\\":\\\"FC133C1E75789\\\",\\\"Size_GB\\\":58.59}}","decoder":{"name":"ossec"},"location":"check_usb_details"},"program":"active-response/bin/usb_prevent_ar_script.exe"}}
2026/08/27 12:51:52 active-response/bin/usb_prevent_ar_script.exe: USB storage enabled (rule ['100016'])
2026/08/27 12:51:52 active-response/bin/usb_prevent_ar_script.exe: Ended
```

The active-response execution log from the Windows 11 Pro endpoint validates the full lifecycle of the automated threat containment workflow.

- **At 12:48:51**, upon triggering rule **100016**, the custom script executes and successfully logs: **USB storage disabled (rule ['100016'])**, instantly modifying the registry to block potential exfiltration.
- **At 12:51:52** (exactly 3 minutes / 180 seconds later), the stateful timeout concludes, triggering the clean-up phase where the script re-enables the storage driver to restore standard endpoint functionality without human intervention.

## Recommendation of detecting or blocking attacks

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As a SOC Analyst, I recommend implementing a comprehensive endpoint threat containment strategy focused on hardware-level monitoring and automated SOAR pipelines to mitigate insider threats and data exfiltration risks

- **Hardware Whitelisting Implementation:** Extend the current script logic to enforce strict USB whitelisting—automatically allowing corporate-approved storage devices via Serial Number while blocking all unauthorized hardware.
  - **Real-time Alert Notifications:** Integrate Postfix mail relays or webhook channels (Slack/Telegram) to ensure the SOC team receives instantaneous high-severity alerts without relying solely on dashboard monitoring.
  - **Proactive EDR Hardening:** Regularly audit Windows Registry keys (USBSTOR) and execution policies across all endpoints to prevent tampering with active response scripts.
  - **Centralized Log SIEM Auditing:** Maintain long-term log retention of Wazuh detection events and active-response execution histories for forensic readiness and compliance reporting.
-